Compiled Problems – Senior Division

1. **00-01 C4 Data Structures**

   Build a binary search tree with the keys LEMONADE (in that order)
   What is the internal path length of the resulting tree?

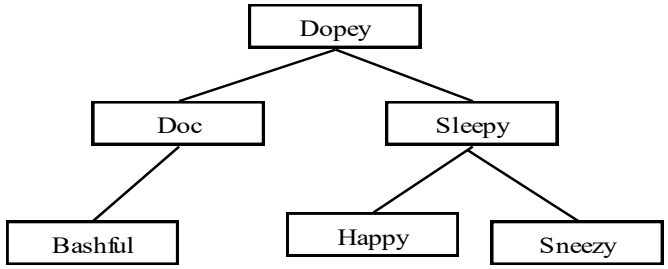
2. **00-01 C4 Data Structures**

   Consider the following sequence of operations on an empty stack.
   A plus sign indicates a pop operation and a letter indicates a push operation of that letter.
   What would be the next item to be popped?

   R A + N + S + O M + + E V E R + + G + L A + D E S + + +

3. **01-02 C4 Data Structures**

   When "Grumpy" is inserted into the following binary search tree,
   what is the new external path length of the tree?



4. **01-02 C4 Data Structures**

   Consider the following operations on an initially empty stack.
   If a POP is now performed, what value would be popped?

   PUSH(B); PUSH(I); PUSH(N); POP(X);
   PUSH(D); POP(X); PUSH(S); POP(X)

5. **02-03 C4 Data Structures**

   Insert the letters THELITTLEMERMAID into an initially empty binary search tree, starting with the letter T and ending with D. How many nodes have 2 children?

6. **02-03 C4 Data Structures**

   Given an initially empty queue and the following list of operations, what would be removed if one additional POP(X) is executed?

   PUSH (M), PUSH (A), PUSH (R), POP (X), PUSH (Y), POP (X), PUSH (P),
   PUSH (O), POP (X), POP (X), PUSH (P), POP (X), PUSH (P), PUSH (I),
   POP (X), PUSH (N), PUSH (S), POP (X), POP (X)

7. **03-04 C4 Data Structures**

Construct a binary search tree for CHICAGOILLINOIS. What is the internal path length of the tree?

8. **03-04 C4 Data Structures**

In the binary search tree above, conduct a post-order transversal of the tree.
List the nodes visited in order.

9. **04-05 C4 Data Structures**

What is the internal path length of the binary search tree for CARNEIGEMELON?

10. **04-05 C4 Data Structures**

Given an initially empty queue and the following operations, what element would be popped next?

PUSH (H), PUSH (E), POP (X), PUSH (R), PUSH (S),
POP (X), POP (X), PUSH (H), PUSH (E), POP (X),
POP (X), POP (X), PUSH (Y)

11. **05-06 C4 Data Structures**

Find the external path length of the binary search tree for: RECURSIVEFUNCTIONS

12. **05-06 C4 Data Structures**

The function REV(X) reverses the contents of a queue and its input order.
What would be the next item removed from the queue generated by the following commands?

PUSH(A), PUSH(B), PUSH(C), REV(X), POP(X), PUSH(D), REV(X), POP(X),
POP(X), PUSH(E), PUSH(F), POP(X), REV(X), POP(X), PUSH(G), REV(X).

13. **06-07 C4 Data Structures**

Find the internal path length of the binary search tree for: KATYTEXAS

14. **06-07 C4 Data Structures**

How many more nodes have one child in the binary search tree for UNITEDKINGDOM than in UNITEDSTATES?

15. **07-08 C4 Data Structures**

Find the internal path length of the binary search tree for: CHARLESDICKENS

16. **07-08 C4 Data Structures**

Define the function REV to reverse the order of the items in the current stack before continuing. What would the next popped element be in the following stack?

PUSH(B), PUSH(L), PUSH(E), POP(X), REV, PUSH(A), PUSH(K), REV, POP(X). POP(X), REV, PUSH(H), POP(X), POP(X), PUSH(O), REV, POP(X), PUSH(S), PUSH(E), REV, POP(X)

17. **08-09 C4 Data Structures**

How many nodes have only 1 child in the binary search tree for: ORDEROFTHEPHOENIX

18. **08-09 C4 Data Structures**

Define the function SW as: switch from a stack to a queue or from a queue to a stack. What would the next popped element be in the following stack?

PUSH(A), PUSH(N), PUSH(I), POP(X), PUSH(M), SW, PUSH(A), POP(X), POP(X), SW, PUSH(T), POP(X), PUSH(I), SW, POP(X), PUSH(O), POP(X), SW, PUSH(N)

19. **09-10 C4 Data Structures**

Find the internal path length of the binary search tree for: CARNEGIEMELLONUNIV

20. **09-10 C4 Data Structures**

Define a new operation **INS(x,N):** insert N into the xth location starting at the pop end. Given an initially empty stack and the following sequence of operations, what would be the next POPPED element?

PUSH(A), PUSH(B), PUSH(C), PUSH(D), INS(2,E), POP(X), PUSH(F), POP(X), INS(4,G), POP(X)

21. **10-11 C4 Data Structures**

What is the internal path length of the binary search tree for the following: BIOHAZARDMATERIALS

22. **10-11 C4 Data Structures**

Define the function SW as: switch from a stack to a queue or from a queue to a stack and REV reverses the elements. What would the next popped element be in the following queue?

PUSH(M), PUSH(U), REV, SW, POP(X), PUSH(S), PUSH(E), SW, REV, POP(X), PUSH(U), REV, PUSH(M), POP(X), SW, PUSH(S)

23. **11-12 C4 Data Structures**

What is the internal path length of the binary search tree for: PHILADELPHIA

24. **11-12 C4 Data Structures**

Given an initially empty queue, what would the next popped element be in the following queue?

PUSH(H), PUSH(E), PUSH(R), PUSH(S), POP(X), PUSH(H), POP(X), POP(X), PUSH(E), POP(X), PUSH(Y), PUSH(P), POP(X), POP(X), PUSH(A), PUSH(R), POP(X), PUSH(K), POP(X)

25. **12-13 C4 Data Structures**

What is the internal path length of the binary search tree for: NORTHCAROLINA

26. **12-13 C4 Data Structures**

Define the function SW as: switch from a stack to a queue or from a queue to a stack. What would the next popped element be in the following queue?

PUSH(M), PUSH(E), PUSH(D), POP(X), SW, PUSH(I), PUSH(T), PUSH(E), POP(X), SW, POP(X), PUSH(R), PUSH(A), POP(X), POP(X), SW, POP(X), PUSH(E), PUSH(A), PUSH(N), SW, POP(X), POP(X), PUSH(S), PUSH(E), POP(X), PUSH(A), POP(X), SW

27. **13-14 C4 Data Structures**

What is the external path length of the binary search tree for: CENTENNIALSTATE

28. **13-14 C4 Data Structures**

What would the next popped element be in the following queue?

PUSH(B), PUSH(L), PUSH(A), POP(X), PUSH(C), PUSH(K), POP(X), POP(X), PUSH(C), POP(X), POP(X), PUSH(A), PUSH(N), POP(X), PUSH(Y), POP(X), POP(X), PUSH(O), PUSH(N), POP(X)

29. **14-15 C4 Data Structures**

Create a binary search tree from the string WALTDISNEY and another binary search tree for DISNEYWORLD. What is the positive difference between the internal path lengths of each tree?

30. **14-15 C4 Data Structures**

Create a binary search tree from the string CINDERELLA. Then add, one at a time, the strings ARIEL, MULAN and BELLE to the original tree. Which one increases the depth of the tree by the most?

31. **15-16 C4 Data Structures**

What is the internal path length of the binary search tree for: LIVEFREEORDIE

32. **15-16 C4 Data Structures**

Define a new command SW as follows: If it is a queue, change to a stack and if it is a stack, change to a queue. Start with an initially empty queue. Perform the following. What is the next element POP'ed?

PUSH(D), PUSH(A), POP(X), SW, PUSH(R), PUSH(T), PUSH(M), POP(X), POP(X), SW, POP(X), PUSH(O), PUSH(U), POP(X), SW, POP(X), PUSH(T), SW, PUSH(H)